

Schematic completion status

PetSafe 100-1339 R03 A | v0.9.14 | 2026-10-09

All catalog entries are drawn on one sheet. The remaining questions are listed explicitly below. This is a partially verified reconstruction, not a completed hardware validation or a routed PCB.

Coverage	Open connections	Values / packages
150 entries; 1 A2 sheet; 20 stock + 4 custom symbol definitions.	8 populated-entry pads + 16 DNP pads. 0 open PIC pads in the model.	44 ceramic and 2 magnetic values unknown; 147/150 footprints assigned.

Power rails - current names

Name	Known path
BATTERY+	Battery positive -> Q1 single pad/native3 (candidate drain); TP14/R43.2 on the raw side.
VSYS	Q1.L/native2 -> U5/MX512H pin4; L1.1 and L2.1 supply sides.
VDD	TP103, PIC20/C32, U5 pin1/VCC, V064/V063/V075/V082. D3.R via V082 is user-confirmed.
VREF	TP104, PIC21/RB0/V026, PIC4/RA2/V035/C39, R21.2/V071. U3.5 and R22 D/left/model2 at 1 ohm.
GND	Battery negative and ground copper.

L1/L2 share the VSYS supply side. Their opposite ends are not shorted together. H_RF_VDD, H_LDO_IN, H_RX_VDD and H_PIR_VDD describe local branches; these are not additional confirmed independent supplies. Rail voltages have not been measured.

Recommended next checks

1. J3/R38, U3.6/R23 B and R22-GND measured; next R23 C to TP6 and both C18 ends.
2. One complete Q3/C10/C11/C12 tuning cell.
3. Specific R41, RF excitation and receiver-output destinations.
4. After routing: frequency-sensitive values and one powered idle/scan session; footprints follow.

Validation and limits

83 modeled net partitions match KiCad 10.0.5. ERC retains 31 findings: 24 open pins, 2 isolated labels and 5 undriven power checks. No artificial NC/power flags hide gaps. Original photo hashes are unchanged. This proves file consistency, not all physical connections.

Isolated modeled nodes: H_R41_FREE, H_D1_FREE. A PIC pin can have a local net and still lack an onward destination.

Completion milestones

Topology complete: Populated components and functional block interfaces have supported connections; candidate functions/polarities are compatible and remaining exceptions individually documented.

Value/package complete: Required values and geometry recovered; exact manufacturer/ratings separated from equivalent design choices. Routed PCB remains a separate deliverable.

Supply and diode gaps

E01 - D2 / marking 4P

Known: Marking 4P. Diode readings support R -> S -> L series-junction behavior in circuit. Photo R=upper-left, L=upper-right, S=single lower. User confirms R-C20 lower 1 ohm; L-ANT2/R46 upper; S-R46 lower. R46 unpopulated. S-ANT2 OL; S-GND 700 kohm, S-VREF 1.32 Mohm. Outer pads show high resistance to VDD/GND.

Uncertain: All three D2 local destinations are resolved. Exact part, ratings and physical package numbering remain unknown; internal series topology is probable from in-circuit tests. Other RX_A_PLUS branches remain inferred. The purpose/value of the empty R46 option is unknown; no resistor is fitted.

Next: No further D2 rail or local continuity sweep is needed. Seek a manufacturer 4P marking/package match consistent with the series-junction measurements; retain the explicit equivalent-circuit assumption if identity cannot be recovered. Do not fit or bridge R46.

E03 - RF supply / Q8 / C6

Known: Q8 centre B2/native5 is GND via V113/V115. V114 is U2.T2/native2 GND. Outer Q8 pads are joined by the earlier low-resistance result. Candidate Q8 P-source/native2 is on filtered VSYS; P-gate/native3 is driven by U2 output4 through R8, while U2 uses VDD.

Uncertain: C6.1 route and Q8 source-supply path. F-P=10 ohm supports a supply-related path but does not establish a direct copper join. Exact Q8 maker remains a candidate. Gate-drive headroom is unverified: a hypothetical source=6V and gate-high=4.5V gives VGS=-1.5V, not guaranteed off for the candidate threshold range. These are illustrative, not measured voltages.

Next: Start with C6's non-ground pad and the L1 output/Q8 supply node; record resistance both ways. Use the existing labelled Q8 photo, not the withdrawn C6-to-output wire. After passive routing, measure VDD plus Q8 source and gate high/low during scanning, all relative to PCB GND; compute VGS. Waveforms require a scope if available. Never place an earth-referenced scope ground on a driven antenna/motor node.

Tuning and local endpoints

E04 - Antenna / five tuning cells

Known: All five PIC control inputs are mapped. Seven capacitor-midpoint comparisons gave OL. User now confirms ANT2 directly to D2.L.

Uncertain: The model uses PNPs with emitter/native2 at GND and base driven through a resistor by nonnegative GPIO. This is not ordinary forward-operated PNP switching. Ground association, pad orientation/type or an RF junction-switching role may be wrong/incomplete. Capacitor-midpoint antenna continuations remain unresolved. ANT2-D2.L is now verified; further tuning/antenna connections remain open.

Next: Finish ONE cell, Q3/C10/C11/C12: map Q3 physical paired/single pads, both terminals of all three capacitors on both sides and the antenna continuation. Start from IMG_2433/IMG_2442 and V107. Compare Q3.L to the established GND point, Q3.R to R13.2, Q3.S to C10.1, and V107 to the relevant C10/C11/C12 junction. Confirm candidate pairs, do not infer them as measured. Use diode tests to constrain transistor type; base identification alone need not distinguish emitter/collector. Then use symmetry for the others. Preserve all seven OL exclusions.

E05 - Button circuit

Known: R41=331 gives 330 ohm. In the current model H_BUTTON reaches PIC28/ICSP_DAT THROUGH R40, not by a direct copper net. S1 common leg pairs remain a normal-switch hypothesis.

Uncertain: R41.1/H_R41_FREE has an unknown destination; its former H_BUTTON_MCU name was not evidence of another MCU connection. A supply, bias or other control destination is possible.

Next: Inspect the short R41 trace in IMG_2438 first. Identify the physical pad that does not reach the S1/R40 button side, then check only candidate nearby VDD, VSYS or TP4/R44 control nodes chosen from the photo. A nonzero resistance is not direct continuity. Confirm S1 common pairs released/pressed. Do not scan all PIC pins or assume a pull-up.

E06 - Local-only PIC / discrete nodes

Known: Every PIC pad has a modeled local net. PIC2 reaches C25/V031; PIC14 reaches TP11; PIC18 reaches TP17. D3.S-to-R42 measured 1.2 ohm; exact resistor end not named, pad2 selected from photo. R42.1 already reaches TP4/PIC25/RB4.

Uncertain: Onward roles of those three nodes, D1.R/H_D1_FREE, C26.1 and TP9.1. Local connectivity is not proof that these are intentionally unused. H_RF_CONTROL has only R6.2/R7.1: its excitation source is absent. H_BAT_SENSE has a series path through R4 to PIC24/RB3, so the missing item is functional interpretation/route confidence, not necessarily another MCU endpoint. H_D3_SIGNAL replaces the misleading H_RX_RF_SENSE name; operating role remains unknown.

Next: Use existing photo/via evidence to select specific endpoints for the local-only PIC nodes, D1.R and H_RF_CONTROL. Review R3-R4-C28 together with PIC24: document a compatible sensing mechanism, without adding an extra sampling wire solely from a net-name assumption.

Other routing and identity gaps

E07 - PIR interface / U6 / J3

Known: J3 pin1 supply previously confirmed; pin3-GND 1 ohm; pin2-R38 K/right/model1 1 ohm, opposite end 10 kohm. Pin2-GND 290 kohm. C2NM supports S-812C33AMC; V067 reaches J3.1/U6A, V063 regulated VDD, and U6 ground is resistance-supported.

Uncertain: U6 VIN/VOUT external branches and NC-pad ties, actual PIR voltage, Q2/R39 and R33 interface branches. Q2.S/TP16 lacks a modeled controller destination. J3 connector ground and R38 entry are now established.

Next: After the receiver input checks, use the Q2/TP16 photo/via route to choose a specific controller endpoint; confirm Q2/R39 and R33 branches if needed. Do not repeat J3 ground or J3-to-R38 tests. Check remaining U6 branches before powered voltages.

E08 - Receiver / bias / Q7

Known: PIC12/RC1-R18 via V053-V070; V064-V075 continuity and V064-V070 11.2 kohm support Q7 supply-switch hypothesis. U3.5 is VREF; R22 D/left is VREF and E/right GND, 10k between. U3.6-R23 B/left 1 ohm, C/right 5.6 kohm; pin6-ground 400 kohm. Prior TP6/C18 fragment is retained separately from pin5.

Uncertain: R23 C/right/model2 onward destination; remaining feedback/coupling and R48.2 VREF hypotheses, Q7 pin roles, receiver output destination. Both R23 ends were 270-289 kohm from U3.5; the old VREF tie remains rejected. U3 pin6 is not directly ground.

Next: Measure R23 C/right to TP6 and to each C18 end. This tests the existing input-filter node as the most plausible continuation; none is assumed short. Keep both C18 ends distinct. Do not repeat pin6-to-R23 or R22-ground checks. Receiver output routing follows.

E09 - Power input / Q1 / rail voltages

Known: BATTERY+ and VSYS are separate. User confirms Q1.L to U5.4 and both L1/L2 supply sides on VSYS. VDD TP is the separate logic rail. R1A matches FMOS3401A marking.

Uncertain: Exact Q1 maker and gate operation, unmeasured rail voltages, and U1 enable/option branches. Reverse-polarity protection is an interpretation, not a powered test.

Next: Cross-check Q1 gate/R44 and U1 enable wiring first. After routing reconciliation, use one powered session on the original hardware/supplies: BATTERY+, VSYS, VDD, VREF, U6 output, Q8 source/gates and both U5 control inputs, idle and during a triggered scan. Receiver power may be switched. No battery/PIC/RF redesign.

E10 - ICSP protection / D6 / J1

Known: D4/D5=5U supports SD05 TVS candidates. D6=G3 has at least two documented Zener candidates. ICSP signal routes are photo-derived. J1 square-pad/VPP end is resolved from IMG_2437 and corroborated by IMG_2439: 1 VPP, 2 VDD, 3 GND, 4 DAT, 5 CLK. Electrical signal destinations are unchanged.

Uncertain: Both D6 pad routes, exact voltage/polarity and candidate diode identities remain open. J1 pitch and lead geometry remain photo-based; its sequential numbering is now resolved.

Next: Measure D6 pads to adjacent PIC/ICSP nodes using the corrected J1 numbering. Do not repeat J1 position identification; position evidence is photographic, not new meter confirmation.

I01 - Candidate identities / transistor pin maps

Known: Primary datasheets and marking evidence are recorded; custom symbols use the selected candidate pin tables. D3 measured rail connections fit the BAV99 clamp candidate; its identity is not newly proven.

Uncertain: Short markings do not uniquely prove manufacturers. Candidate choices and some B/E/C or MOSFET source/drain assignments remain qualified. Western alternatives are redesign options, not identities of fitted parts.

Next: Use electrical pad behavior and package/marketing evidence for specific candidates. Do not request sharper photos where the existing marking is already legible.

I02 - LED / motor connector

Known: V127 reaches the photo-selected LED common pair; the two PIC LED drives are mapped. MX512H OUTA/OUTB map to the modeled motor connector.

Uncertain: LED internal polarity/colour mapping and J5 physical A/B lead polarity remain provisional.

Next: Check LED pad pairs in diode mode and record emitted colour. Match J5 wire positions to U5 outputs by continuity.

Values, mechanics and scope

V01 - Unmarked passive values

Known: C5 is 470 uF / 16 V; 38 fitted resistors have nominal marked values. R8 full-body dimensions support 0603 and C11 dimensions support 1206.

Uncertain: Unmeasured fitted ceramic values and L1/L2 type/value remain listed by the component audit. Voltage rating, dielectric, resistor tolerance and power rating are separate evidence gaps.

Next: After one tuning cell topology is verified, prioritize its effective capacitance, RF shaping and receiver filters before bypass values. C11/C12 presently share both modeled nodes: an in-circuit reading is their effective network value, not individual values. Record frequency, mode, amplitude and in-circuit/isolated condition. Separate individual values only when needed for the BOM; do not assign an invented split or blanket 100nF.

M01 - Package geometry / PCB reproduction

Known: Most entries have candidate stock footprints; R8 and C11 have user dimensions. J1 sequential numbering is corrected, but exact land-pattern dimensions remain photo-based.

Uncertain: C5, S1 and LED1 footprints are blank. Other families, connector pitch, test-pad drills, switch lead span and exact board geometry are partly photo-based.

Next: Defer C5/S1/LED1 dimensions until the main signal paths are coherent; then measure can/lead spacing, switch body/lead span and LED pad pitch. Assigned families are not verified land patterns.

D01 - Empty options

Known: 16 open pads belong to unpopulated options. R46 is now fully mapped: upper to D2.L/ANT2, lower to D2.S, with no resistor fitted.

Uncertain: Optional-pad routing is incomplete; values and intended uses may remain unknowable without another board variant.

Next: Record option copper where visible; defer measurement unless it affects a populated net.

F01 - Function / internal layers / daughterboard

Known: The one-sheet schematic has a native/model connectivity check and retained photo/via evidence; see the current validation counts.

Uncertain: No powered functional validation, actual layer stack or routed KiCad PCB has been created. The PIR daughterboard internal circuit is not reconstructed. MCU firmware behavior and operating timing are not established.

Next: Finish priority electrical gaps first. Then test defined functions/voltages and decide whether physical PCB reproduction and daughterboard reverse engineering are needed.

Every unrecovered ceramic value

C1, C2, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C43, C44, C45, C46, C47, C48

C5 is already 470 uF / 16 V. In-circuit LCR readings may include parallel components and semiconductor paths; record frequency, mode and whether a lead was isolated. Geometry cannot establish capacitance, dielectric or voltage rating.

Closed issues retained as history

E02 - User identifies D3.L=GND and D3.R=VDD through V082. Move only D3.R off the hypothetical receiver supply. D3.S-to-R42 measures 1.2 ohm (resistor end not stated; photo selects pad2). Signal function remains with E06 and candidate identity with I01.

PIC connections

Physical package pins. All have modeled local nets; the evidence column distinguishes confirmed endpoints from photo-derived or incomplete branches. Use the separate numbered photo guide for physical locations.

Pin	Function	Local destination / evidence
1	RE3/MCLR	J1.1/VPP at square-pad end; position photo-supported; D6 route missing
2	RA0	C25 signal pad / V031; onward destination unknown
3	RA1	R28 + R29 + TP7; user meter / visible trace
4	RA2	VREF TP via V035; user confirms via-to-TP; C39.1 local photo route
5	RA3	V036-V108 to R14; user continuity report
6	RA4	V037-V102 to R15; user continuity report
7	RA5	V039-V097 to R16; user continuity report
8	VSS	GND / V041 photo anchor
9	RA7/OSC1	Y1 oscillator; photo-derived
10	RA6/OSC2	Y1 oscillator; photo-derived
11	RC0	V047-V091 to R11; user continuity report
12	RC1	R18.1 via V053-V070; user-confirmed; VREF link explicitly rejected
13	RC2	TP3 via V049; user-confirmed; existing R10 branch photo-derived
14	RC3	TP11; visible route; onward destination unknown
15	RC4	V022-V016 to U5 pin2 / INA; user-confirmed
16	RC5	V023-V019 to U5 pin3 / INB; user-confirmed
17	RC6	R13; user meter / visible trace
18	RC7	TP17; visible route; old R13 guess withdrawn
19	VSS	GND / C32 / V045 photo anchor
20	VDD	V042 to board VDD TP user-confirmed; C32.1 local photo route
21	RB0	VREF (TP104), V026 and V071/R21.2; user-confirmed
22	RB1	TP2 / R31; user meter / visible trace
23	RB2	TP1 / R30; user meter / visible trace
24	RB3	R4; photo-derived; sensing role remains inferred
25	RB4	TP4 / V017 / R42.1 / R44.1; user continuity; C41 photo-derived
26	RB5	TP10 / R35; user meter / visible trace
27	RB6/CLK	ICSP CLK / D4 / R32; photo-derived
28	RB7/DAT	ICSP DAT / D5 / R40; photo-derived

Every remaining open pad

Populated entries: C26.1, C6.1, D6.1, D6.2, R23.2, TP9.1, U6.L1, U6.L2.

Empty options: C42.1, C42.2, C49.1, C49.2, J6.1, J6.2, Q9.S, R36.1, R36.2, R47.1, R47.2, U6A.L, U6A.R2, U7.1, U7.2, U7.3.

U6.L1/L2 are candidate internal NC pins 4/5; external ties remain unknown. Photo pad S means single pad, not necessarily source. Current completion checklist includes closure criteria and the full per-component audit. Sources: reconstruction.json, remaining_work.json, via_audit.json, v099_net_changes.json, native validation and the preserved earlier measurement records.